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BBM434 END OF TERM PROJECT
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WATCHLOG

A SMART GPS SYSTEM FOR
HIKERS AND BIKERS

BIKER

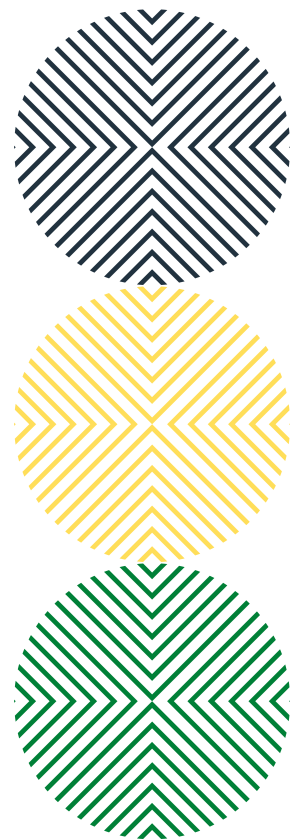


WHAT IS IT?

Outdoor enthusiasts need a reliable way to track their performance, specifically speed, altitude, and duration while on the move. WatchLog Timer was designed to meet this need, providing a smart, real-time GPS dashboard dedicated to nature sports.

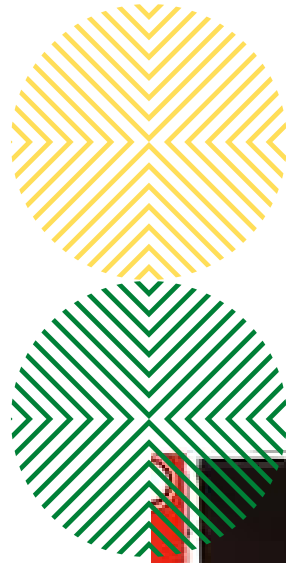


No one wants their phone to die at the peak of a mountain. Unlike smartphones, which burn through battery rapidly when tracking GPS, WatchLog Timer is built for endurance. It offers over 20 hours of continuous runtime, saving your phone's battery for emergencies. It delivers professional-grade tracking reliability at a fraction of the cost.

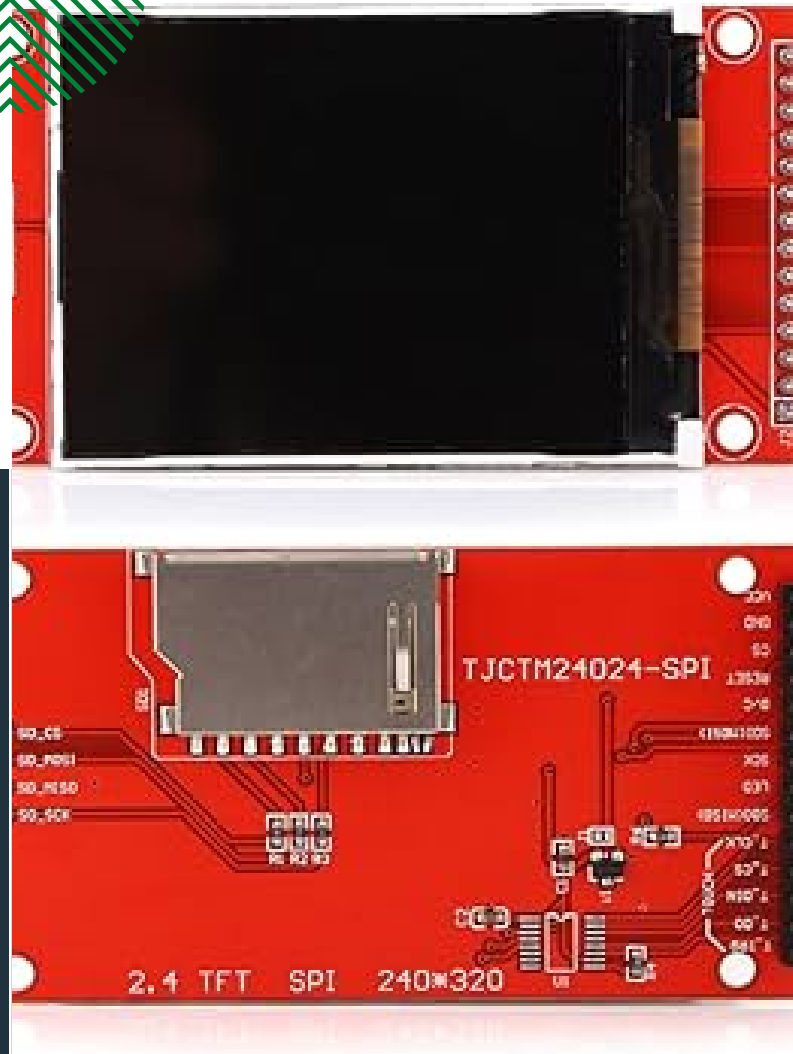


**BUT ALREADY WE HAVE
PHONES AND SMART
WATCHES!**



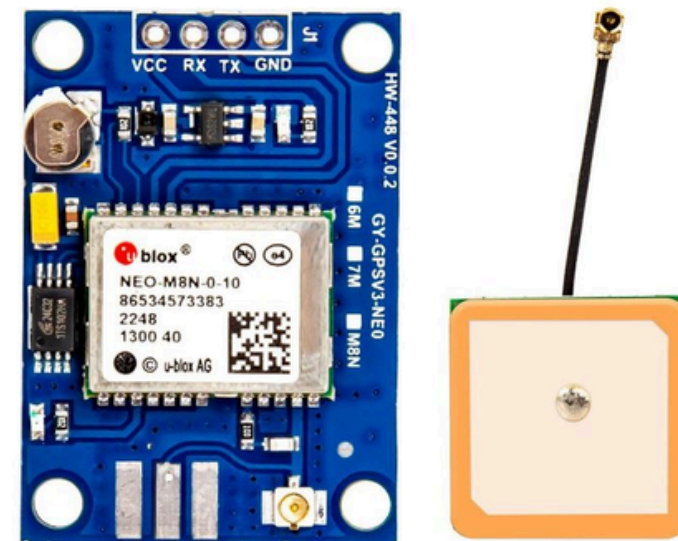


THE MAIN COMPONENTS



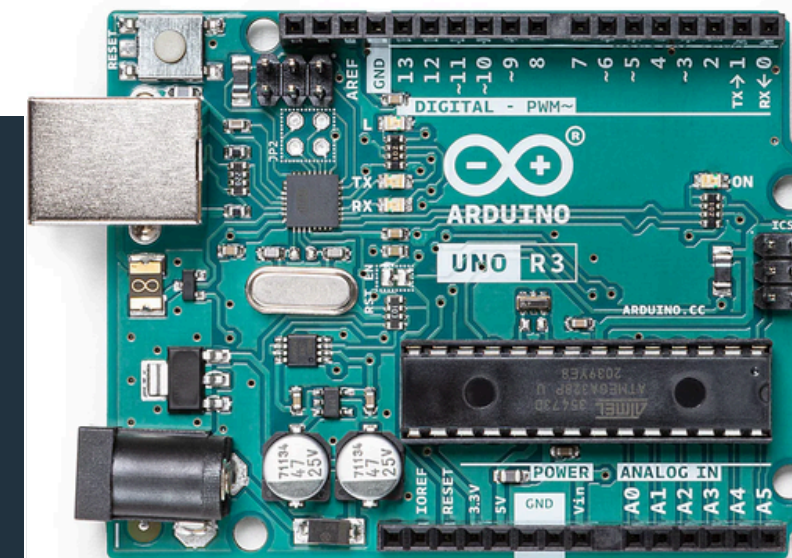
ILI9341 TFT SPI LED

A TFT LED uses SPI with touch support.



NEO-M8N GPS

GPS module that periodically gets coordinates from USA/Russian satellites.



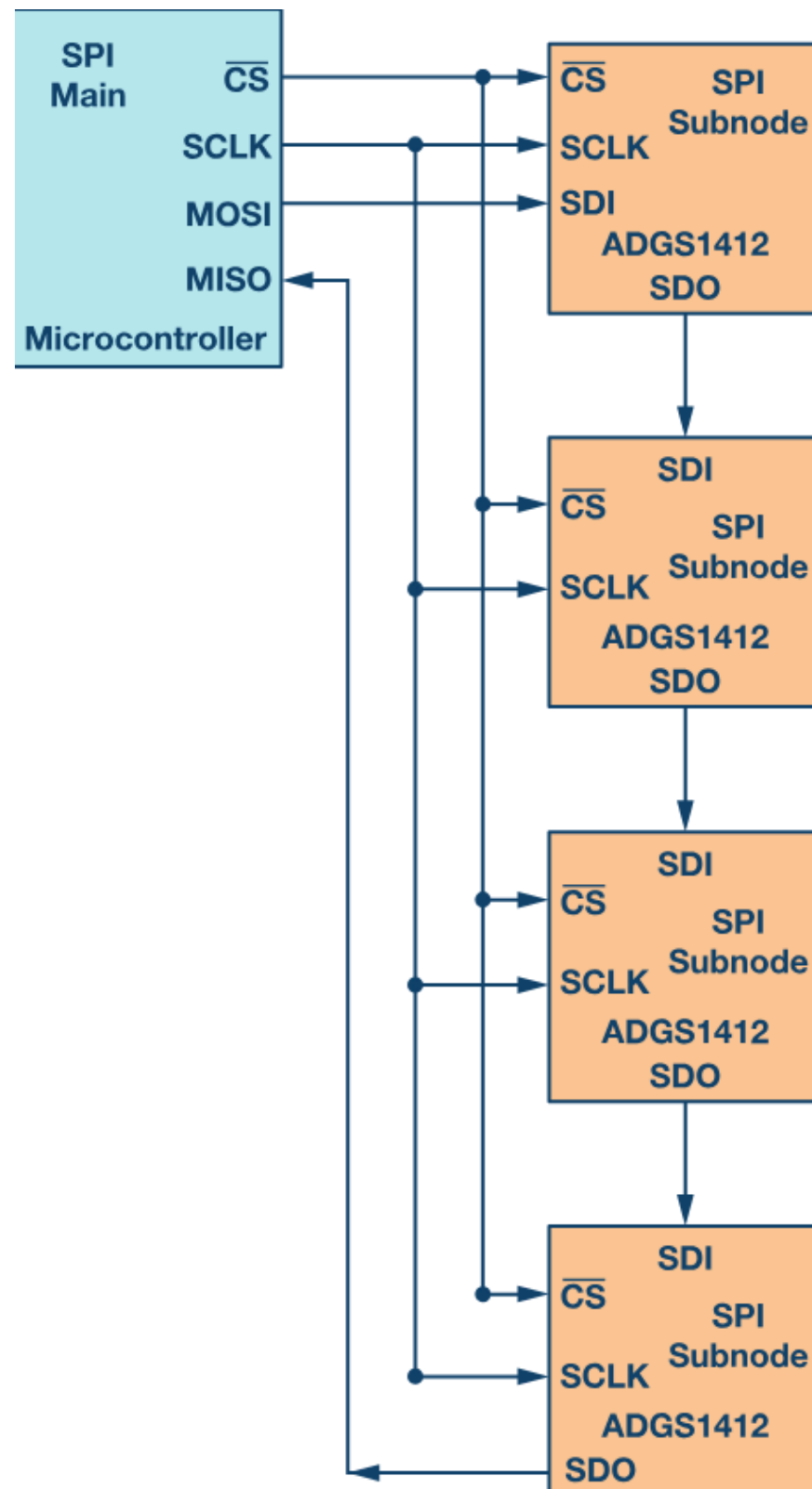
ARDUINO UNO

MCU card.



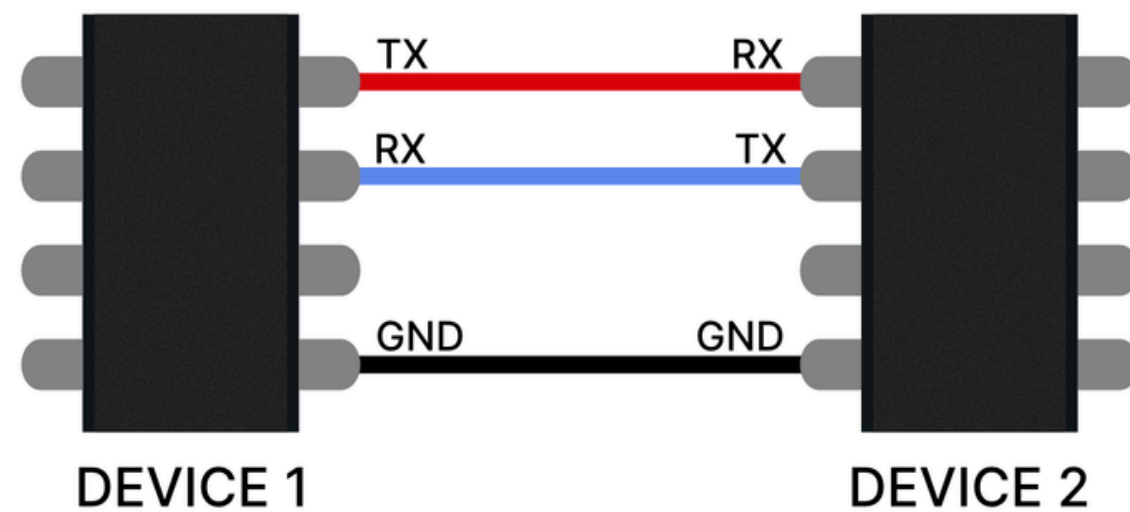
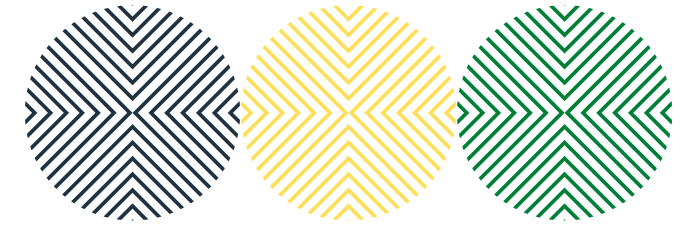
8GB SD CARD

To save logs and coordinates.



TECHNICAL STORY

Optimized Shared-Bus Architecture "Engineered with a high-efficiency Shared SPI Interface, allowing the TFT Display and SD Storage to operate seamlessly on a single data bus. The system utilizes precision Chip Select (CS) Arbitration to intelligently manage traffic, ensuring zero-latency switching and guaranteeing data integrity during simultaneous visualization and logging tasks."



GPS Communication Interface: UART, The positioning module utilizes a Universal Asynchronous Receiver-Transmitter (UART) interface for reliable data streaming.

- Protocol: Adheres to the standard NMEA 0183 protocol, transmitting ASCII-encoded geospatial data.
- Configuration: Operates via a Cross-Coupled RX/TX Architecture, allowing the microcontroller to parse real-time coordinate streams without blocking the main execution loop.
- Baud Rate Stability: configured for a precise 9600 bps data rate to ensure error-free asynchronous framing."

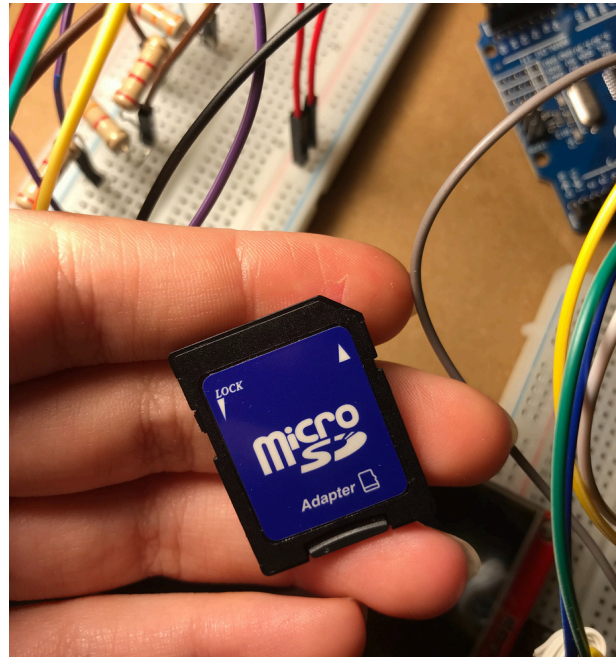
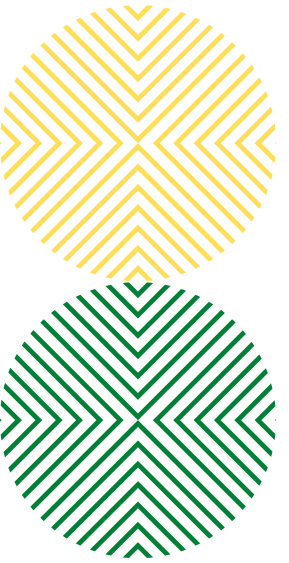


FUTURE PLANS

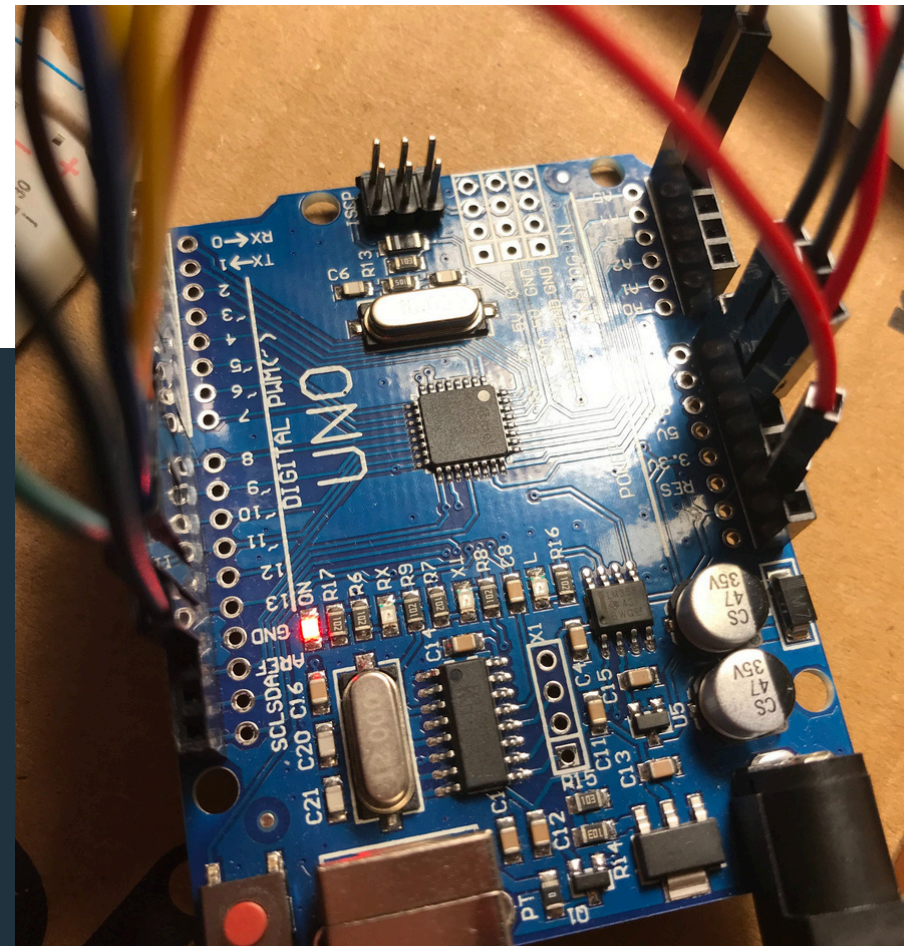
Moving forward, we plan to evolve WatchLog Timer into a fully connected device. Our next phase involves integrating GSM and Wi-Fi modules, allowing users to broadcast their live location for safety and sync data wirelessly. To support these demanding features and introduce a touch-enabled interface, we will also be upgrading the core processor to a high-performance microcontroller.



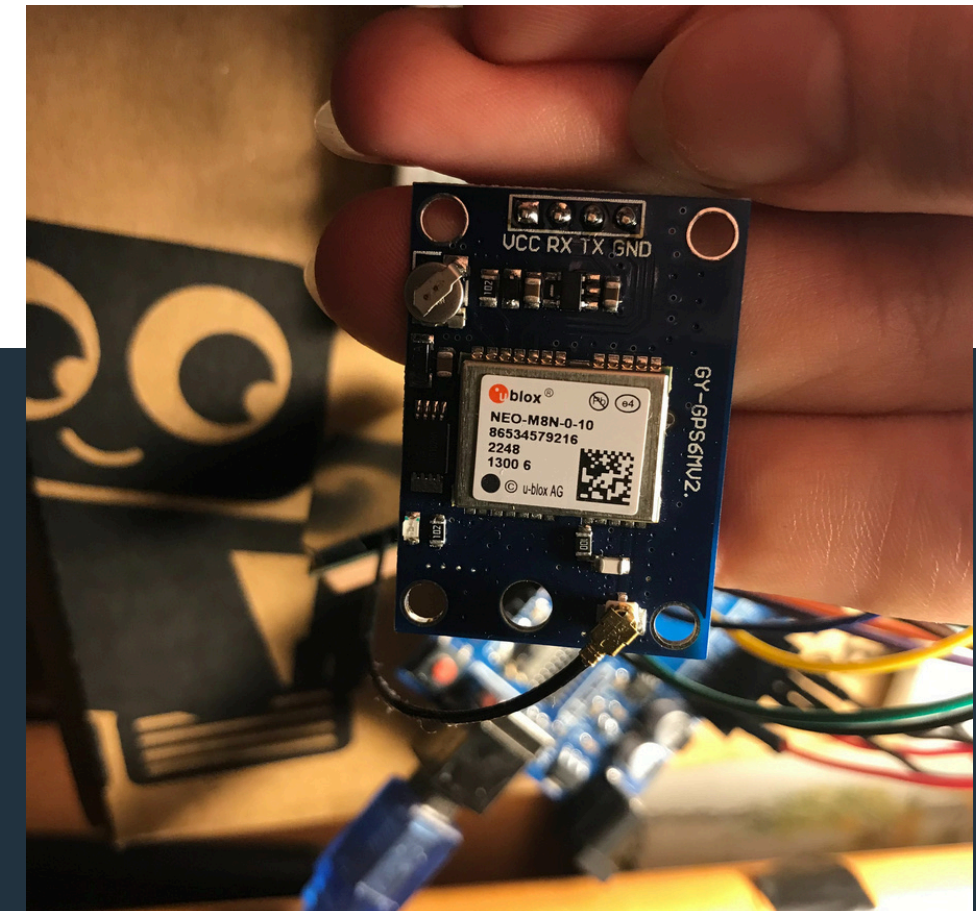
CIRCUIT PHOTOS



TFT LED



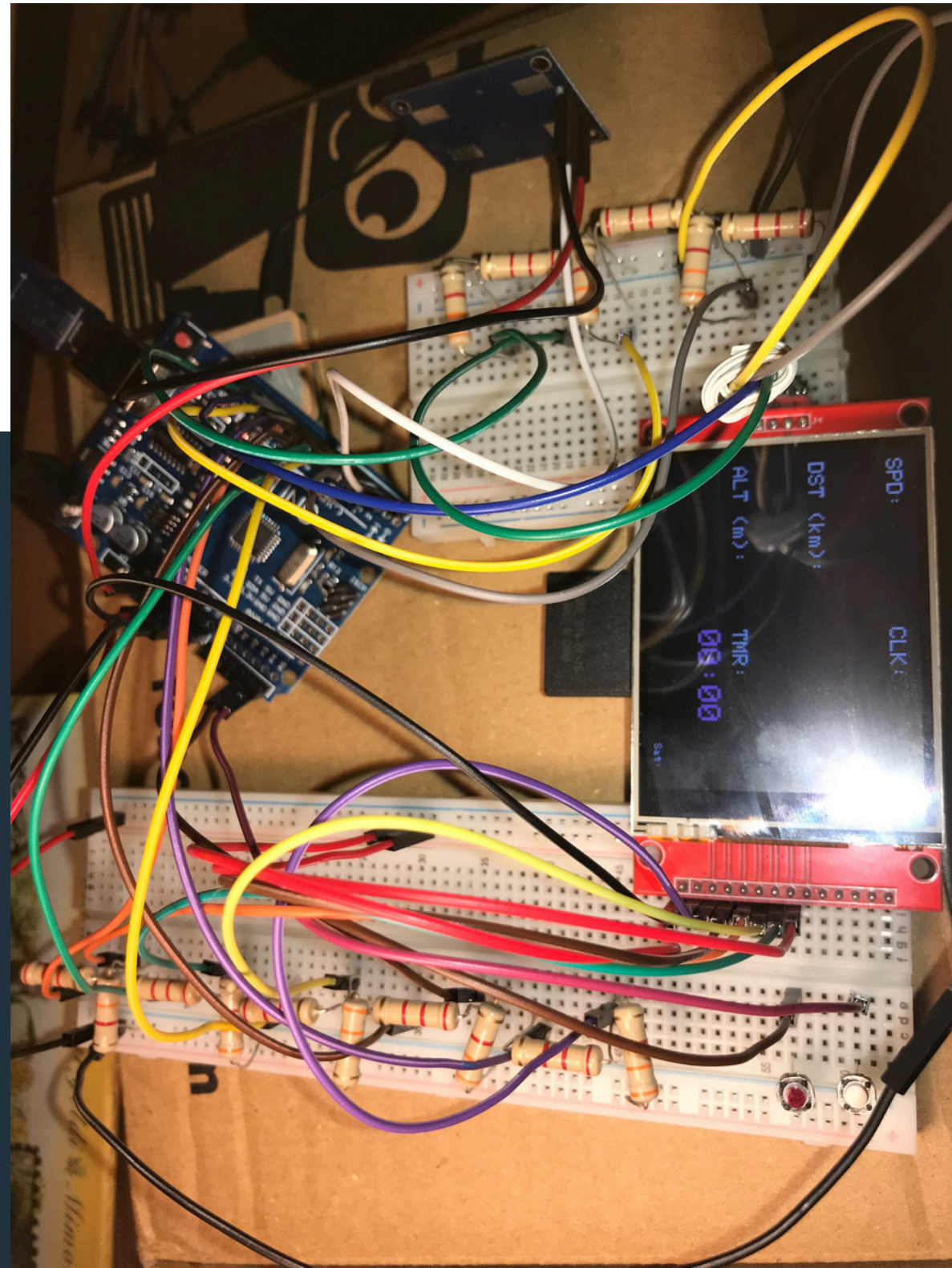
Arduino UNO



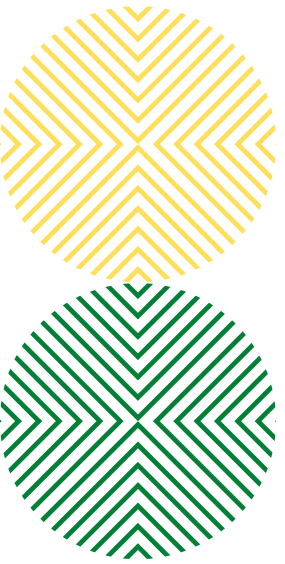
NEO M8N GPS

CIRCUIT PHOTOS

WatchLog Biker Prototype
<https://drive.google.com/file/d/1alhN0uM1p5hHx9gxp7EXpwT9AD3JfgHy/view?usp=sharing>



GPS working version
<https://drive.google.com/file/d/1qggynPeKnpYDTxYAVNMhSBdJIKelFWf3/view?usp=sharing>



THANK YOU
FOR LISTENING

